

TOULOUSE FRANCAZAL AP, France

WMO: 076310

Lat: 43.5403N

Lon: 1.3727E

Elev: 535

StdP: 14.41

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.7	28.5	19.6	15.0	29.8	22.9	17.6	31.8	25.1	47.2	21.6	47.5	5.0	280	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	92.7	70.3	89.0	69.1	86.0	68.2	72.6	86.6	71.2	84.4	69.8	81.9	7.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	105.7	77.8	66.9	101.0	76.8	65.6	96.5	75.6	36.5	86.5	35.3	84.5	34.1	82.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	18.5	16.3	23.3	98.7	3.0	3.9	21.1	101.5	19.4	103.8	17.7	105.9	15.5	108.8
			22.0	75.1	3.5	1.3	19.5	76.1	17.5	76.8	15.5	77.6	13.0	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	43.1	44.4	50.3	55.0	61.4	69.2	72.7	72.5	66.9	60.1	49.6	44.4
	DBStd	12.30	6.44	6.84	5.94	5.38	5.84	6.07	5.40	5.41	5.46	6.59	6.71	6.72
	HDD50	772	225	174	69	14	2	0	0	0	0	8	87	193
	HDD65	3504	678	577	454	303	140	26	6	4	41	174	463	638
	CDD50	3524	12	16	80	163	356	575	704	696	506	322	74	20
	CDD65	785	0	0	0	2	30	152	245	235	97	24	0	0
	CDH74	6828	0	0	2	48	276	1334	2229	2101	760	78	0	0
	CDH80	2525	0	0	0	4	56	481	899	871	209	5	0	0
Wind	WSAvg	7.3	7.2	8.0	8.5	8.4	7.7	7.3	7.3	6.7	6.5	6.8	6.9	6.7
Precipitation	PrecAvg	25.7	2.1	1.9	2.0	2.6	2.9	2.4	1.7	1.9	2.0	2.1	2.1	2.0
	PrecMax	36.2	5.7	3.7	4.6	6.7	7.1	10.4	7.3	5.9	5.9	6.4	5.7	6.6
	PrecMin	16.0	0.1	0.2	0.0	0.4	0.8	0.3	0.1	0.4	0.0	0.4	0.1	0.1
	PrecStd	4.8	1.2	1.0	1.1	1.4	1.3	1.6	1.2	1.1	1.4	1.5	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	65.1	72.9	79.7	85.8	94.8	96.2	99.3	89.4	80.3	69.5	61.6
		MCWB	54.8	53.3	56.2	61.4	65.4	69.9	70.1	71.2	68.2	64.4	60.1	55.1
	2%	DB	57.0	60.2	67.5	74.1	80.7	88.9	92.2	93.0	85.6	75.9	65.5	58.6
		MCWB	52.3	51.5	55.0	59.1	64.0	69.4	69.9	70.5	66.5	64.0	58.3	53.1
	5%	DB	54.7	57.1	63.7	69.7	76.6	85.5	88.5	88.9	82.0	72.8	62.3	56.4
		MCWB	50.9	50.0	53.1	57.1	63.0	68.2	69.0	69.5	65.1	62.7	57.1	52.0
	10%	DB	52.5	54.3	60.3	65.7	72.8	81.8	85.2	85.0	78.4	70.0	59.4	54.1
		MCWB	49.4	49.2	51.9	55.4	61.1	66.5	68.1	68.3	64.4	61.8	55.3	50.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	55.1	58.6	62.8	68.0	73.5	74.2	74.7	70.8	67.6	61.5	56.2
		MCDB	58.6	62.0	67.6	76.0	80.3	88.2	89.6	89.4	84.0	75.5	67.5	60.3
	2%	WB	53.3	52.8	56.1	60.3	65.8	71.4	72.4	72.7	69.0	66.0	59.4	54.0
		MCDB	56.2	58.3	64.4	71.0	77.6	85.2	86.4	86.6	79.9	72.1	64.0	57.6
	5%	WB	51.6	51.2	54.4	58.3	63.9	69.6	71.1	71.1	67.5	64.7	57.6	52.6
		MCDB	54.3	55.6	61.7	67.8	73.8	82.2	84.0	84.0	77.7	70.3	61.4	55.9
	10%	WB	49.6	49.7	52.9	56.3	62.1	67.7	69.5	69.6	65.9	63.2	55.8	51.0
		MCDB	52.1	53.8	58.9	64.0	70.8	79.0	81.1	81.6	75.1	68.7	58.8	53.8
Mean Daily Temperature Range	5% DB	MDBR	11.9	14.2	16.1	16.7	16.8	18.5	19.4	20.1	19.3	16.0	12.7	12.7
		MCDBR	14.4	19.3	22.7	24.5	24.4	25.2	26.5	27.8	24.9	19.2	16.2	15.1
		MCWBR	10.2	12.0	12.1	12.2	10.9	9.6	8.8	9.1	9.4	9.2	10.0	10.6
	5% WB	MCDBR	12.4	16.7	18.2	21.5	20.5	22.2	21.9	22.6	19.6	14.6	13.8	13.6
		MCWBR	9.3	11.4	10.4	11.3	10.1	9.4	8.5	8.5	8.0	7.4	9.1	9.9
Clear-Sky Solar Irradiance	taub	0.320	0.336	0.376	0.398	0.407	0.418	0.411	0.411	0.379	0.366	0.339	0.319	
	taud	2.476	2.411	2.305	2.250	2.280	2.283	2.305	2.315	2.389	2.430	2.460	2.479	
	Ebn at Noon	257	271	272	275	275	271	272	268	268	257	246	246	
	Edh at Noon	23	29	36	42	41	42	40	38	33	28	23	21	
All-Sky Solar Radiation	RadAvg	460	758	1151	1488	1694	1944	1926	1722	1379	919	543	428	
	RadStd	40	84	105	156	123	136	135	104	69	74	59	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page